Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-based Neuromuscular Disorder Management Platform: Personalized Pyridostigmine dose optimization with multimodal patient monitoring using voice analysis, PRO, and biometric data			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio-Medial	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Interventional Therapeutic Devices	Therapeutic Devices & Diagnostic Devices	Intelligent Interpretation Systems
Concept of Proposed Technology	The proposed technology is a specialized Software as a Medical Device (SaMD) framework designed to address the gap between outpatient visits in Myasthenia Gravis (MG) management. Effective myasthenia gravis management is critically important because even small fluctuations in neuromuscular strength can rapidly escalate into life-threatening respiratory or bulbar crises, making continuous, precise, and timely monitoring essential for patient safety and quality of life.			

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	- **Multimodal Data Fusion**: Synchronizes voice-based fatigue markers, Patient-Reported Outcomes (PRO), and medication history with clinical reference data (e.g., VOG-SFI, antibody panels). - **Predictive AI Engine**: Utilizes pharmacometric (PopPK/PD) modeling and time-series analysis to forecast 24-hour fatigue curves and recommend optimal dosing intervals. - **Safety & Decision Support**: Implements real-time safety checks and consultation alerts to detect exacerbation risks or medication side effects (infection, metabolic changes) between outpatient visits.
Rationale for Joint R&D Project	Developing a clinically reliable and regulatory-ready Software as a Medical Device (SaMD) for myasthenia gravis (MG) requires tightly coordinated expertise across clinical neurology and patient cohort operations, advanced AI and pharmacometric modeling, and medical-grade software engineering. This joint R&D consortium aligns these capabilities into a single execution team: clinicians ensure clinical relevance and validation, the Canadian research institute leads voice biomarker and multimodal AI development, and Nanum Space drives the clinical-grade data pipeline and PK/PD-informed dose optimization core for Korea-based deployment and commercialization. Together, the partners can deliver continuous between-visit monitoring, personalized pyridostigmine optimization, and safety-focused decision support that would be difficult for any single organization to achieve alone. Beyond this complementarity, the collaboration is further motivated by the following unmet needs: **Objective Dosing Tools**: Medication (Pyridostigmine) titration often relies on empirical trial-and-error; objective digital biomarkers can support safer and more consistent dosing decisions. **Unmonitored Intervals**: Patients are typically seen every 2-3 months, during which daily symptom fluctuations may go untracked and timely interventions can be delayed. **Interoperability and Standardization**: Establishing a standardized, FHIR-aligned data pipeline for neuromuscular disorders supports domestic data governance and enables broader adoption across healthcare institutions.
Key Words	1. Myasthenia Gravis (MG) 2. Personalized Dose Optimization 3. Voice-based Digital Biomarkers 4. Software as a Medical Device (SaMD) 5. Multimodal AI Fusion

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	- **AI Engine for Dose Optimization**: Develop algorithms to predict fatigue curves and optimize Pyridostigmine timing. - **Digital Biomarker**: Create a voice-based fatigue index to quantify bulbar symptoms (dysarthria, nasalization). - **Exacerbation Prediction**: Build models to detect early warning signs of disease worsening in advance. - **Clinical Utility Validation**: Collect multi-center longitudinal data to verify the safety and efficacy of the platform.			
R&D Target Specifications	Target specifications are framed as high-level, differentiating deliverables that leverage Nanum Space's strengths in clinical-grade data standardization, pharmacometric modeling, and SaMD execution: (1) MG-specific voice digital biomarker that captures fatigue-related speech signatures and demonstrates clinically meaningful alignment with standard MG assessments. (2) Multimodal longitudinal data integration (voice, PRO, medication logs, wearable biometrics, and clinical anchors) using interoperable, FHIR-aligned structures suitable for multi-site deployment. (3) PK/PD-informed dose optimization module that supports individualized pyridostigmine timing recommendations and provides clinician-interpretable rationale. (4) Safety guardrails and alert logic to flag potential under-treatment, over-medication, and early exacerbation risk between outpatient visits. (5) Patient and clinician user experience designed for routine outpatient care, including low-burden data capture and actionable dashboards. (6) Evidence-generation workflow and documentation set aligned with SaMD regulatory expectations (risk management, clinical evaluation, cybersecurity, and usability). (7) Scalable cloud/edge deployment architecture enabling secure rollout across Korean clinical sites and future expansion to other neuromuscular conditions.			

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R&D Plan	**Phase 1 (Year 1) - Data Platform & Initial Model Development:** - M1-6: Design and develop patient mobile application (iOS/Android) MVP with core data collection features including medication logging, PRO entry, and voice recording. - M7-12: IRB approval, patient recruitment initiation (N=20-30), voice dataset collection and labeling, initial voice fatigue model development (target accuracy: 70%+). **Phase 2 (Year 2) - Multimodal AI Integration & Pilot Validation:** - M13-18: Voice model enhancement (accuracy 80%+), multimodal fusion model development, dose optimization algorithm implementation with safety guardrails. - M19-24: Silent mode pilot (retrospective AI vs actual dosing comparison), open mode pilot with patient feedback, cumulative cohort N=50-100. **Phase 3 (Year 3) - Prototype Completion & Regulatory Preparation:** - M25-30: AI engine finalization, complete patient application and clinician dashboard prototypes. - M31-36: Clinical validation data analysis, regulatory documentation package preparation (technical file, clinical evaluation report, risk management file). Final cohort N=120-150.

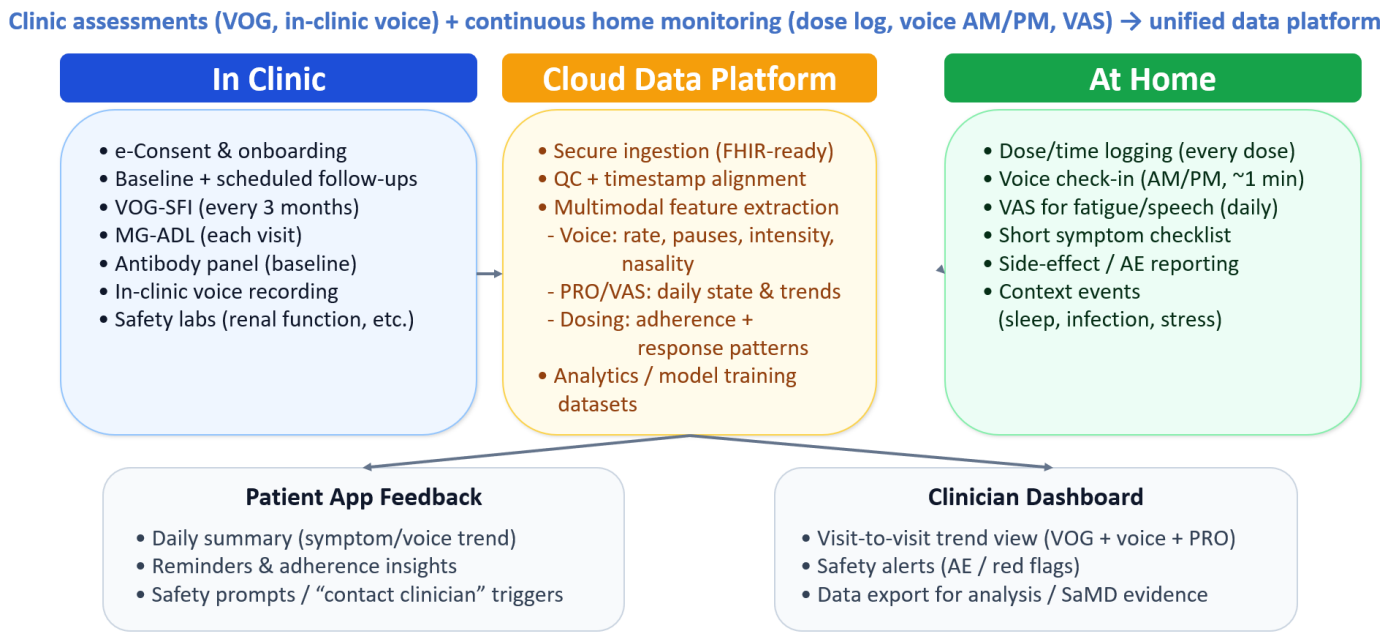

Figure 1. Clinical Data Collection Workflow

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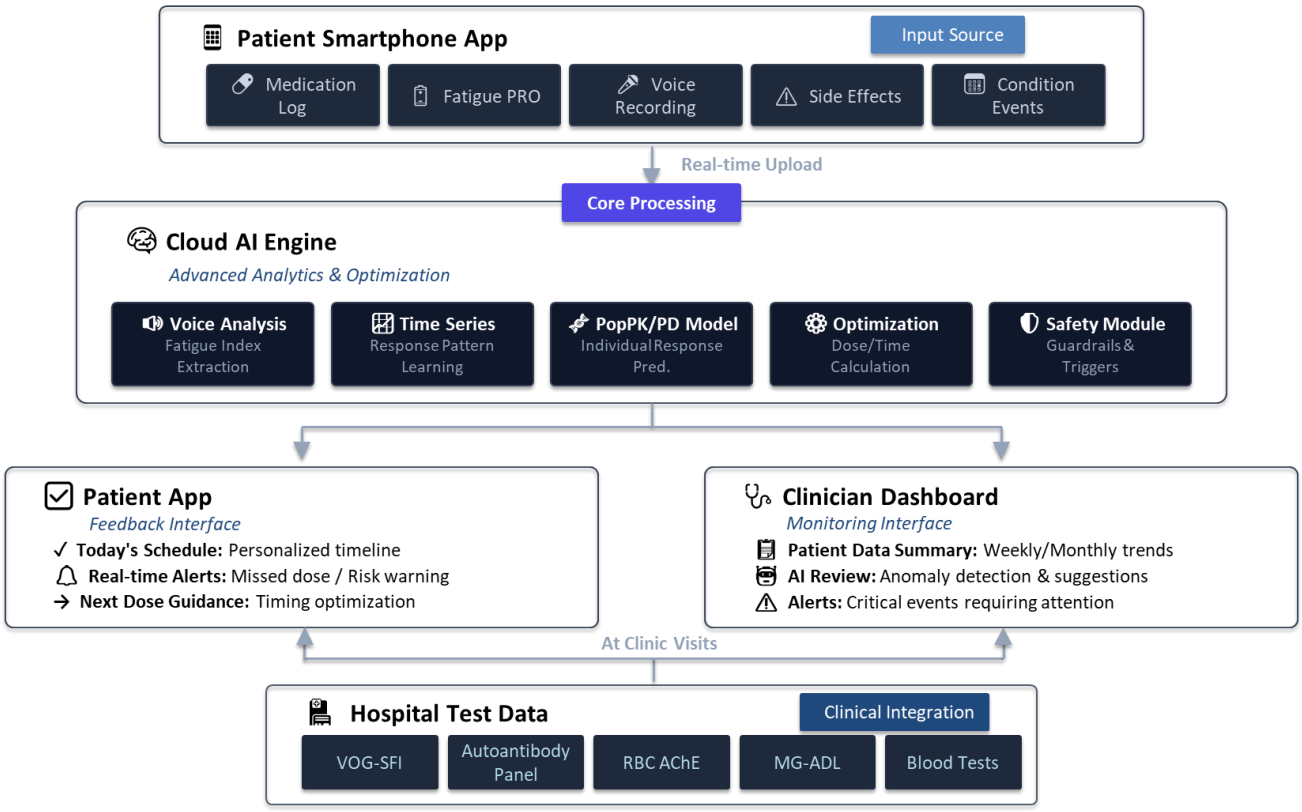

Figure 2. Data Flow Architecture

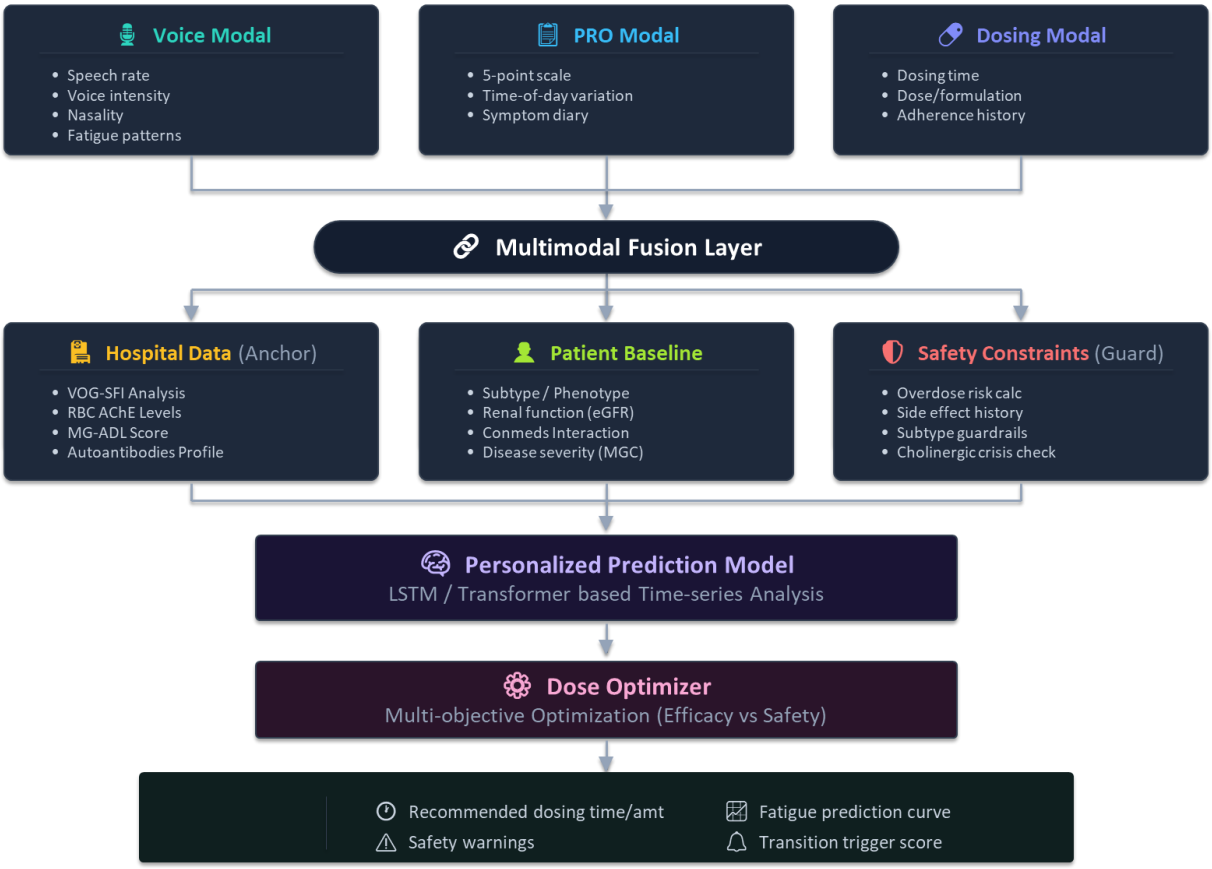

Figure 3. Multimdal AI Engine Architecture

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Assessment	Baseline (M0)	M3	M6	M9	M12	Frequency / Notes
In-clinic assessments						
VOG-SFI	●	●	●	●	●	Every 3 months
MG-ADL (PRO)	●	●	●	●	●	Each visit
In-clinic voice recording	●	●	●	●	●	Standard tasks (≈3–5 min)
Antibody panel (AChR / MuSK / LRP4)	●					Baseline only
Safety labs (renal function, CBC, etc.)	●		●		●	Baseline + as clinically indicated
At-home monitoring (between visits)						
Medication log (dose, time, formulation)	▬	▬	▬	▬	▬	Every dose (≈30 sec)
Voice check-in (smartphone)	▬	▬	▬	▬	▬	Twice daily (AM/PM, ≈1 min)
VAS score (fatigue / speech / chewing)	▬	▬	▬	▬	▬	Daily (or pre/post-dose)
Condition & AE reporting	▬	▬	▬	▬	▬	Daily + event-triggered (sleep, infection, stress)

● Clinic visit item ▬ Continuous home monitoring

Figure 4. Assessment Schedule (Example of a 12-Month Follow-Up)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**The Canadian Research Institute partner shall be responsible for:** - Application Development: Design, develop, and deploy iOS/Android patient mobile application and web-based clinician dashboard with secure cloud infrastructure. - AI/ML Model Development: Lead development of the voice-based fatigue analysis model including acoustic feature extraction, MG-specific features (nasality, fatigue progression patterns), and deep learning classification models. - Multimodal Fusion Architecture: Design and implement the multimodal data fusion framework integrating voice biomarkers, PROs, medication history, and clinical anchors using attention-based mechanisms. - Model Optimization: Develop edge-deployable model versions for on-device inference, implement privacy-preserving learning techniques for multi-site deployment. - Technical Consultation: Provide guidance on scalable AI architecture for future multi-center expansion. - Resources to be Provided: GPU computing cluster, ML engineering team, graduate research assistants, and access to relevant acoustic analysis tools and datasets.
Expected R&R of Korean SME Partner	**The Korean SME partner shall be responsible for:** - Data Infrastructure: Build data collection platform, FHIR-compliant data standardization pipeline, and secure API integrations. - PK/PD Modeling: Develop population pharmacokinetics model for pyridostigmine and implement the dose optimization engine. - Clinical Coordination: Coordinate with Korean clinical sites for IRB submissions, patient recruitment, clinical data collection (VOG-SFI, autoantibody panels, blood tests), and data quality control. - Regulatory Strategy: Lead SaMD regulatory pathway planning and documentation preparation for MFDS and Health Canada submissions. - Project Management: Overall project coordination, milestone tracking, budget management, and consortium communication. **The Korean clinical research site (hospital) shall be responsible for:** - Patient recruitment: Enrollment based on inclusion/exclusion criteria. - Clinical Testing: Conducting clinical examinations and tests (e.g., VOG-SFI, autoantibody panels, MG-ADL assessments). - Monitoring: Clinical safety, including adverse event reporting. - Consultation: Providing clinical interpretation and advisory on results, such as trigger definitions for subtypes.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Anticipated results include: (1) an integrated prototype (patient app + clinician dashboard) with end-to-end multimodal monitoring and dosing decision support; (2) a multi-center, longitudinal multimodal dataset and standardized data pipeline enabling continued model refinement and future extension; (3) a regulatory-ready technical documentation and evidence package aligned with MFDS and Health Canada SaMD pathways; (4) dissemination and technology transfer outputs (e.g., publications, conference presentations, and partner-ready commercialization assets). Expected impact includes improved continuity of MG care by reducing between-visit blind spots, enabling earlier clinician intervention, and supporting more consistent pyridostigmine management. From an industrial standpoint, the project strengthens Nanum Space and Korean partners' capabilities in interoperable clinical data infrastructure, pharmacometric AI, and SaMD regulatory readiness, creating an exportable platform that can be expanded to other neuromuscular disorders.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization